# Darshan V

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 8.13 |
| RV PU COLLEGE | Class XII (Karnataka School Examination and Assessment board) | 2024 | 95.67% |
| ST PAUL'S ENGLISH SCHOOL | Class X (Indian Certificate of Secondary Education) | 2022 | 97.40% |

## PROJECTS

**Credit Risk Scoring for Thin-File Borrowers**

**GitHub | Live Demo**

*Tech Stack: Python, LightGBM, scikit-learn, pandas, SHAP, Streamlit*

- Designed and built a data pipeline integrating **6 relational tables** (307,511 records, 44M+ behavioral records) into a unified feature store, handling table joins, missing data, and feature aggregation across multiple source tables
- Engineered **111 features** through systematic aggregation of 5 supplementary tables (late payment ratios, bureau debt ratios, credit utilization) combined with 5 custom domain features
- Trained and compared Logistic Regression, Random Forest, and LightGBM using 5-fold stratified cross-validation; benchmarked class weighting, SMOTE, and threshold tuning on an 8% default rate dataset
- Applied SHAP value decomposition for model interpretability; deployed as a live Streamlit application achieving **ROC-AUC of 0.7755**

**Customer Lifetime Value (CLV) Prediction for Retail Customers**

**GitHub**

*Tech Stack: Python, PyTorch (LSTM), LightGBM, scikit-learn, pandas, Gradio*

- Built an end-to-end data pipeline on the Online Retail II dataset (**1M+ transactions, 4,023 repeat customers**), engineering time-aware train/predict splits with a fixed cutoff date to prevent data leakage
- Applied log1p target transformation to handle heavily right-skewed revenue data, and designed and compared three modeling architectures -- RFM + LightGBM, sequence-based LSTM, and a hybrid model
- Ran a 5-seed evaluation to control for initialization variance, catching and fixing a model-checkpointing bug; final baseline achieved **$R^2$ 0.31, MAE £613**, an evidence-based architecture decision over the more complex deep learning approach
- Deployed an interactive Gradio dashboard with a full 11-iteration experiment log for reproducibility

**SQL Business Analytics -- Chinook Database**

*Tech Stack: PostgreSQL, SQL*

- Designed and queried a multi-table relational database (6+ tables) in **PostgreSQL**, using joins, CTEs, and window functions to process and transform raw transactional data into analysis-ready structures
- Built queries answering 8 distinct data questions on revenue, retention, and sales performance, demonstrating relational data modeling and multi-step SQL logic
- Ranked customer spend within regional segments and calculated aggregate retention metrics across the full customer base, uncovering revenue patterns driven by transaction volume rather than transaction value

## RELEVANT COURSES

**Programming & Math:** Intro to Programming · Data Structures and Applications · Intro to Statistics & Probability · Linear Algebra

**Entrepreneurship** : Intro to Entrepreneurship · Innovation Management · Finance for engineers

## SKILLS

**Languages:** C, C++, Python, Java, SQL

**Data Engineering:** Pandas, NumPy, PostgreSQL, SQL

**ML & Data Science:** scikit-learn, LightGBM, PyTorch, SHAP, Feature Engineering

**Dev Tools:** Git, GitHub, Jupyter/Google Colab

## POSITION OF RESPONSIBILITIES

**Sponsorship Head, MILAN (Inter-Hostel Fest, IIT Hyderabad)**

- Led **10 coordinators**, securing **6 brand partnerships** for MILAN, IITH's annual fest, worth **Rs. 8L** in sponsorships so far
- Increased sponsorship value by **15%** YoY through targeted outreach across **4 new sponsor categories** for MILAN
- Negotiated sponsorship terms and coordinated partner deliverables, aligning sponsor expectations with event constraints

**Public Relations Head, Cultural Council -- IIT Hyderabad**

- Led a **7-member PR team**, managing publicity and outreach for **10+ cultural and literary clubs** under the Cultural Council
- Redesigned outreach strategy targeting incoming freshers, increasing club sign-ups by **40%** compared to the previous year
- Served as the key link between the Cultural Council and external stakeholders, coordinating outreach across student bodies.

**Head of Filmmaking, Behind the Lens (BTL)**

- Directed IITH's official Inter IIT Culturals 8.0 film, leading **15 filmmakers** and managing a **Rs**. **2L budget** end-to-end
- Mentored **20+ students** through hands-on workshops, teaching cinematography, editing & writing through practical projects